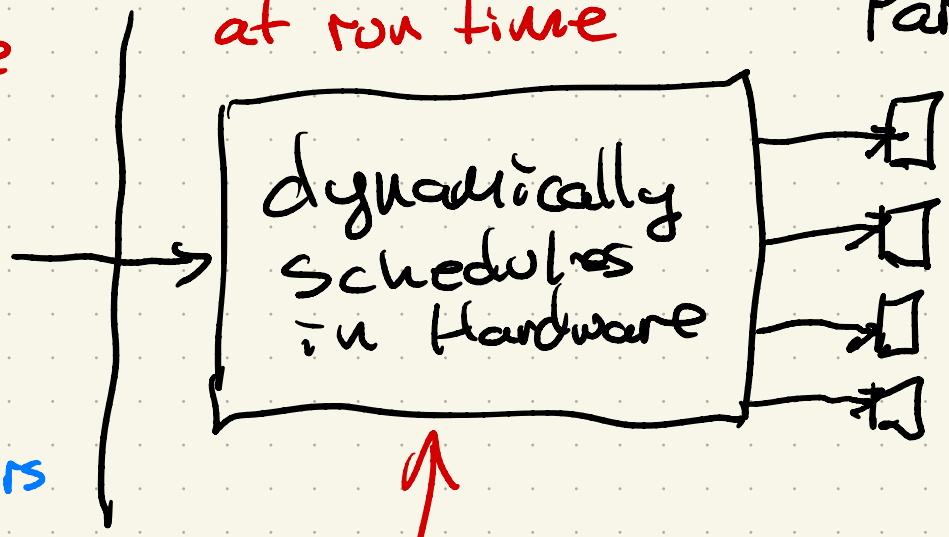
Extend HW0 deadline by 2 weeks 4/15

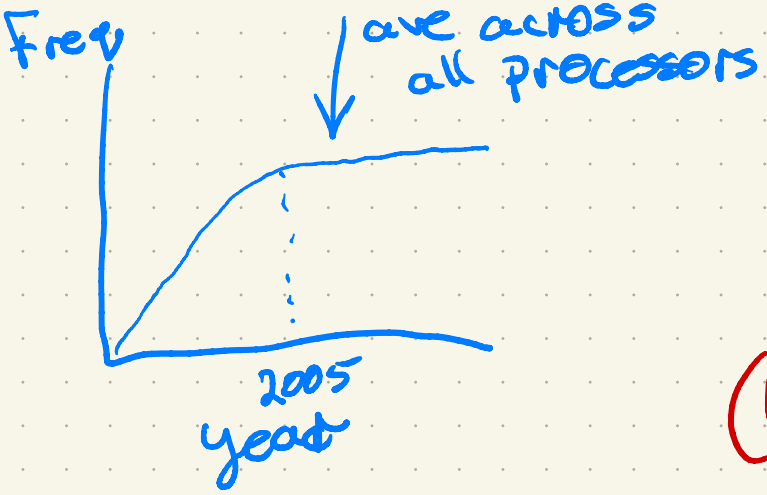
Superscalar

before run time

Unscheduled code

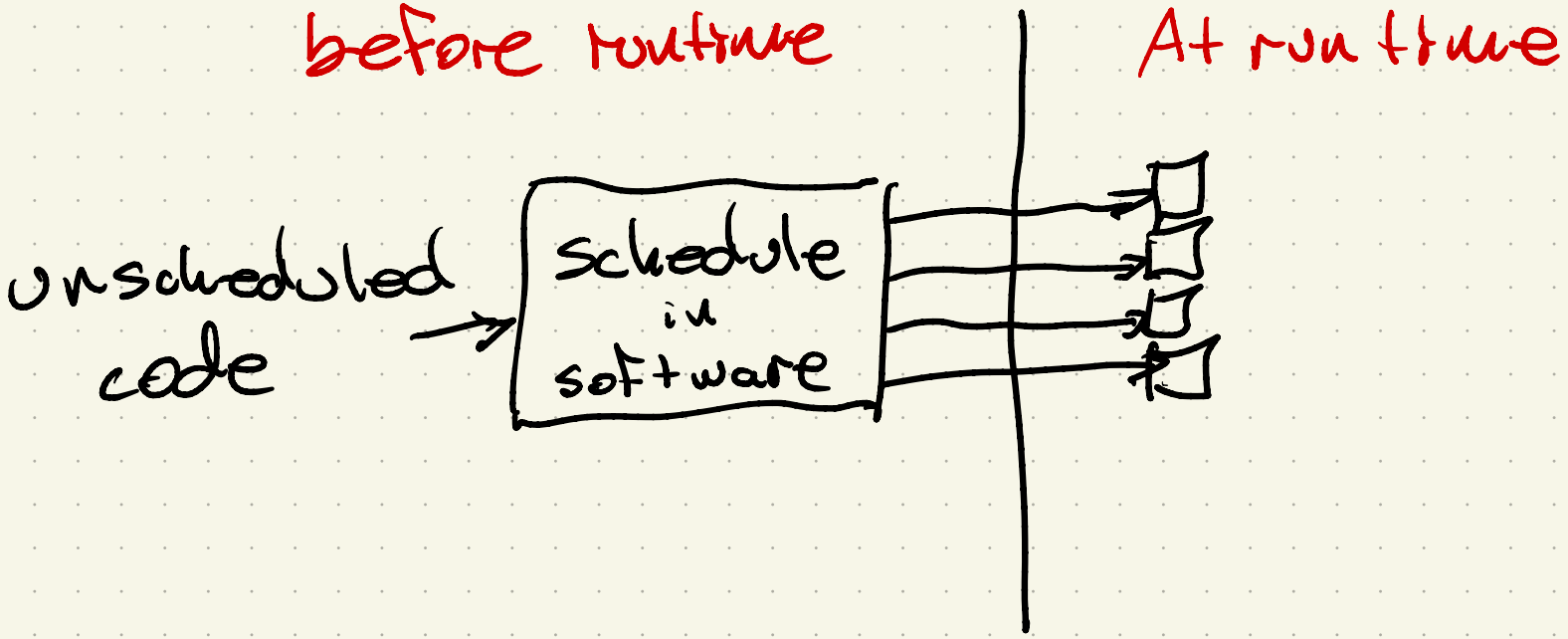


Parallel execution



Drawbacks:

① complex — increased CT and/or increased power



"word" of operations/cycle

Very long instruction word

VLIW

"featuring" parallelism

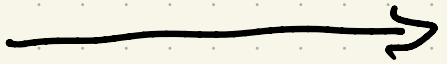
Back end of compiler

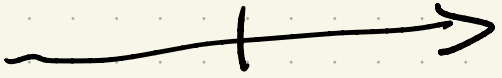
Data dependance graph

Nodes: instructions (operations)

Arcs: Directed

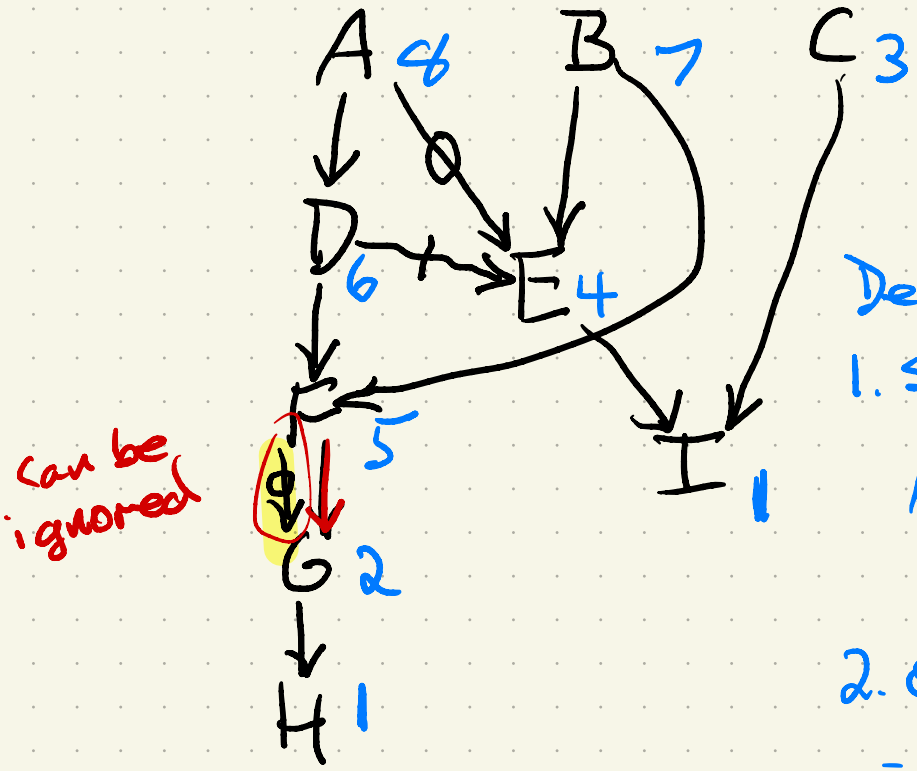
From x to y means x must begin execution before y

Pure/flow 

Anti-dep. 

Output dep. 

A: LW R2, X
B: LW R3, Y
C: LW R4, Z
D: Add R1, R2, #1
E: Mul R2, R3, R3
F: Mul R5, R1, R3
G: Add R5, R5, #1
H: SW R5, X
I: Add R6, R7, R4



Assign priority:

Dependence height

1. start at bottom of DDG
   Assign ops their FU latencies

2. op x has height = Max of all successor ops + its latency

A
B
D
F
E
C
G
H
I

Scheduling Heuristics

$O(n^2)$

Horizontal NOPs

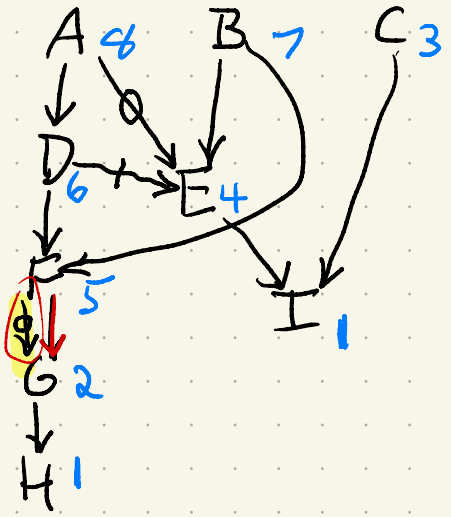
Add			Mul			Load / Store	
d	s	s	d	s	s	d/s	addr
						R2	X A
						R3	Y B
R1 D	R2	#1				R4	Z C
			R5	A	R3 F		
			R2	R3	R3 E		
R5	R5	1 G					
R6	R2	R4 I				R5	X H

~~A~~ ~~B~~ ~~D~~ ~~I~~ ~~E~~ ~~C~~ ~~G~~ ~~H~~ ~~I~~

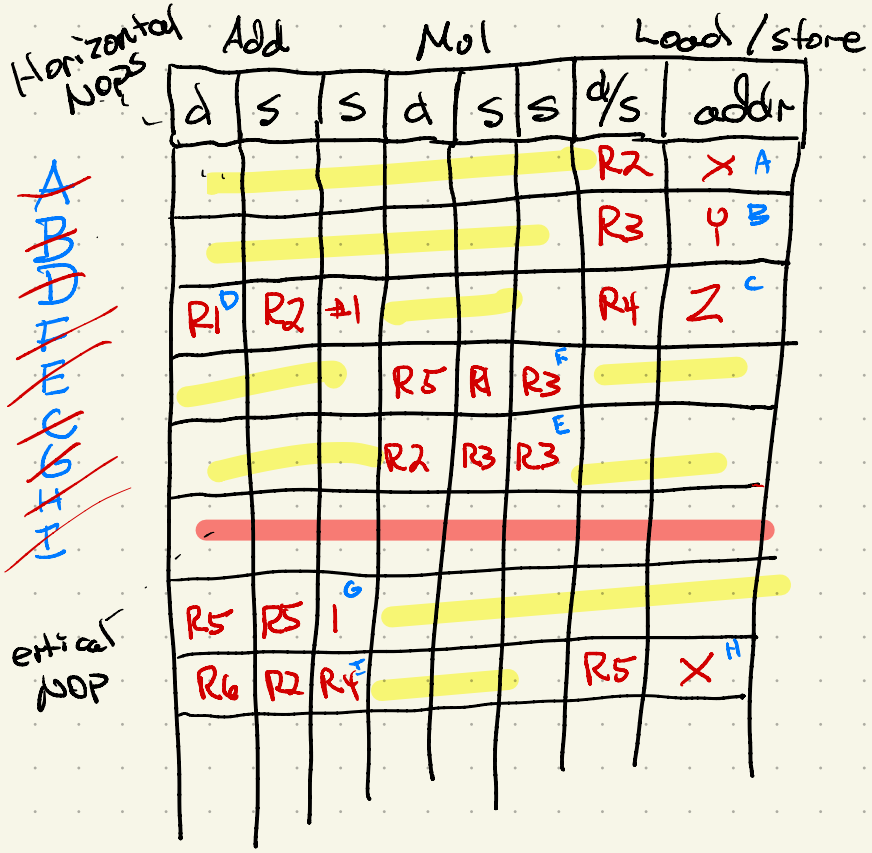
Vertical NOP

"operations"

A: LW R2, X
B: LW R3, Y
C: LW R4, Z
D: Add R1, R2, #1
E: Mul R2, R3, R3
F: Mul R5, R1, R3
G: Add R5, R5, #1
H: SW R5, X
I: Add R6, R2, R4



"smart compiler, dumb machine"



32 bits

Opcode	Dest	Src	Src	Tail?	Parse
LW	R2	X	-	1	0
LW	R3	Y	-	1	0
ADD	R1	R2	#1	0	0
LW	R4	Z	-	1	0
MUL	R5	R1	R3	1	0
MUL	R2	R3	R3	1	1
ADD	R5	R5	#1	1	0
ADD	R6	R2	R4	0	0
SW	R5	X	-	1	0

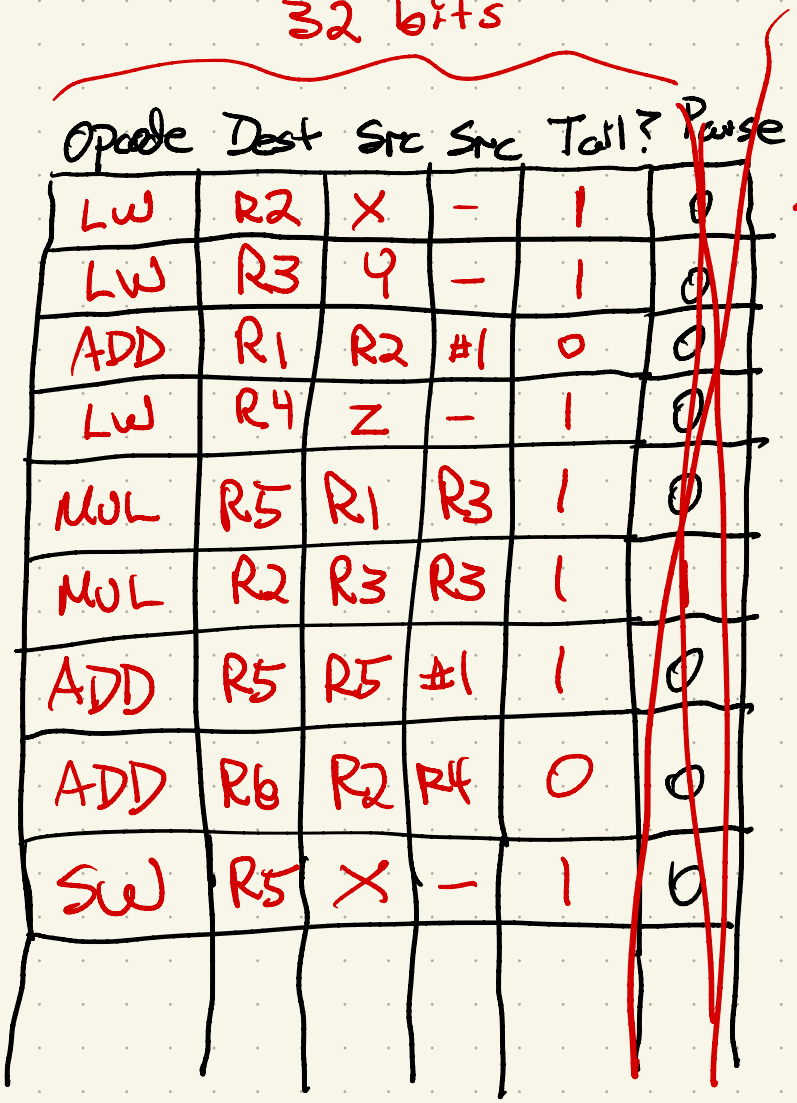
Add FICO at the "row" (VLIW) level

VLIW & OPS

Multiop & op

Groups & instructions

Molecules & atoms



set R2 busy
R3
R1
R4

clear R2 busy
clear R3 busy

⋮

Itanium

List scheduling:

1) Build Data dependence graph
2) sort nodes into a linear order based on priority assignment
3) Insert operations into blank schedule cycle by cycle